Literature Review Template

1. Review Metadata

Review title:
`[Short title of the literature review]`

Research area:
`[e.g. Context graphs for AI research agents]`

Prepared by:
`[Agent / human / team]`

Date:
`[YYYY-MM-DD]`

Library topic ID:
`[topic_id]`

Related direction:
`[direction_id]`

Status:
`Draft / Under review / Complete / Needs update`

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2. Review Objective

This literature review investigates:

> `[State the research area or problem being reviewed.]`

The purpose of the review is to understand the current state of the field, identify dominant methods, map open problems, compare existing approaches, and identify a research gap that could support a new contribution.

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3. Research Questions for the Review

This review is guided by the following questions:

1. `[What is currently known about the topic?]`
2. `[What methods are commonly used?]`

3. `[What datasets, benchmarks, or evaluation protocols are standard?]`
4. `[What limitations appear repeatedly across the literature?]`
5. `[Where is the field saturated?]`
6. `[What gap could become a viable research direction?]`

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4. Search Scope

Sources searched

- arXiv
- Google Scholar
- GitHub
- OpenML
- Hugging Face
- Conference proceedings
- Workshop papers
- Reputable technical blogs
- Internal Library records

Search terms

```
[term 1]
[term 2]
[term 3]
[term 4]
```

Inclusion criteria

Papers or sources were included if they:

- directly relate to `[research topic]`
- introduce or evaluate a relevant method
- discuss a relevant benchmark or dataset
- identify limitations, gaps, or future work
- are recent or foundational
- have sufficient provenance and trust metadata

Exclusion criteria

Sources were excluded if they:

- are unrelated to the research direction
- lack clear provenance
- are low-trust or quarantined by Mimir
- make unsupported claims

- duplicate already-indexed work
- are too shallow to inform research direction

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5. Field Overview

High-level summary

`[Write a concise overview of the field. What is the topic? Why does it matter? Why is it active now?]`

Main research clusters

Cluster	Description	Representative papers	Maturity
`[Cluster 1]`	`[Description]`	`[Papers]`	Emerging / Mature / Saturated
`[Cluster 2]`	`[Description]`	`[Papers]`	Emerging / Mature / Saturated
`[Cluster 3]`	`[Description]`	`[Papers]`	Emerging / Mature / Saturated

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6. Key Papers

Paper	Year	Contribution	Method	Dataset / Benchmark	Limitations	Relevance
`[Paper 1]`	`[Year]`	`[Contribution]`	`[Method]`	`[Dataset]`	`[Limitations]`	High / Medium / Low
`[Paper 2]`	`[Year]`	`[Contribution]`	`[Method]`	`[Dataset]`	`[Limitations]`	High / Medium / Low
`[Paper 3]`	`[Year]`	`[Contribution]`	`[Method]`	`[Dataset]`	`[Limitations]`	High / Medium / Low

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7. Method Landscape

Common methods

Method	Used by	Strengths	Weaknesses	Notes
`[Method 1]`	`[Papers]`	`[Strengths]`	`[Weaknesses]`	`[Notes]`
`[Method 2]`	`[Papers]`	`[Strengths]`	`[Weaknesses]`	`[Notes]`
`[Method 3]`	`[Papers]`	`[Strengths]`	`[Weaknesses]`	`[Notes]`

Method relationships

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[Method A] extends [Method B]
[Method C] is commonly used as a baseline for [Method D]
[Method E] addresses limitation [X]
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8. Dataset and Benchmark Landscape

Dataset / Benchmark	Used for	Used by	Strengths	Risks / Limitations	Approved by Mimir
`[Dataset 1]`	`[Task]`	`[Papers]`	`[Strengths]`	`[Risks]`	Yes / No / Pending
`[Dataset 2]`	`[Task]`	`[Papers]`	`[Strengths]`	`[Risks]`	Yes / No / Pending

Evaluation standards

Common evaluation metrics include:

- `[Metric 1]`
- `[Metric 2]`
- `[Metric 3]`

Common experimental patterns include:

- `[Pattern 1]`
- `[Pattern 2]`
- `[Pattern 3]`

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9. Claims in the Literature

Claim	Supporting papers	Contradicting papers	Evidence strength	Notes
`[Claim 1]`	`[Papers]`	`[Papers]`	Weak / Medium / Strong	`[Notes]`
`[Claim 2]`	`[Papers]`	`[Papers]`	Weak / Medium / Strong	`[Notes]`

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10. Limitations and Open Problems

Recurring limitations identified in the literature:

1. `[Limitation 1]`
2. `[Limitation 2]`
3. `[Limitation 3]`

Open problems:

1. `[Open problem 1]`
2. `[Open problem 2]`
3. `[Open problem 3]`

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11. Saturated Areas

The following areas appear crowded or difficult to differentiate:

Area	Reason it may be saturated	Risk
`[Area 1]`	`[Reason]`	Low / Medium / High
`[Area 2]`	`[Reason]`	Low / Medium / High

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12. Research Gaps

Potential gaps identified:

Gap 1: `[Gap title]`

Description:
`[Explain the gap.]`

Why it matters:
`[Explain why the gap is important.]`

Evidence from literature:
`[List supporting papers or patterns.]`

Potential contribution:
`[Describe what a new paper could contribute.]`

Risk:
`[Explain why this gap may be hard to pursue.]`

Gap 2: `[Gap title]`

Description:
`[Explain the gap.]`

Why it matters:
`[Explain why the gap is important.]`

Evidence from literature:
`[List supporting papers or patterns.]`

Potential contribution:
`[Describe what a new paper could contribute.]`

Risk:
`[Explain why this gap may be hard to pursue.]`

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13. Recommended Research Direction

Based on the literature, the most promising direction is:

> `[Recommended research direction]`

Novelty rationale

`[Explain how this differs from prior work.]`

Why now?

`[Explain why this is timely.]`

Why this lab can pursue it

`[Explain why LabFoundry has the right data, agents, tools, or architecture.]`

Main risks

- `[Risk 1]`
- `[Risk 2]`
- `[Risk 3]`

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14. Library Links

Related papers:
`[paper_ids]`

Related methods:
`[method_ids]`

Related datasets:
`[dataset_ids]`

Related claims:
`[claim_ids]`

Related experiments:
`[experiment_ids]`

Related requests:
`[request_ids]`

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15. Final Review Verdict

Verdict:
`Proceed / Needs more review / Not promising / Saturated`

Reason:
`[Explain the verdict.]`

Recommended next artifact:
`Research proposal / Field brief / Additional review / Dataset investigation`